

**YEAR 8 SCIENCE
CHEMICAL SCIENCES
REVISION - TEST 1**

1. Look at p. 230 Science Quest 8 and use it to answer the following about the development of our current model of the atom. For each person, list what he or she found about the atom.

Democritus (500 BC):

Dalton (1803):

Rutherford (1911):

Bohr (1931):

Chadwick (1932):

2. Describe the structure of an atom. Draw a labelled diagram to help your explanation, listing the component particles and their charges.

3. Define the following terms.

Element:

Compound:

Mixture:

Atomic number:

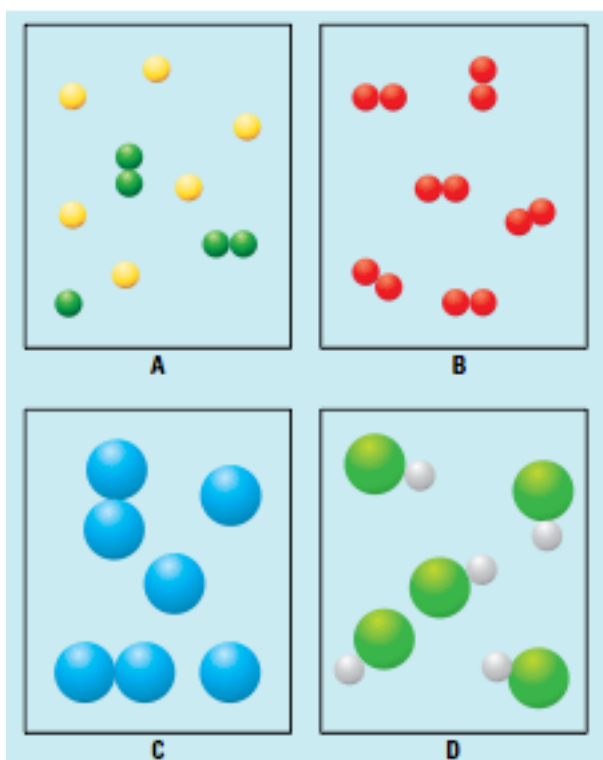
Atomic mass:

4. Which of the diagrams below represent:

(a) elements.

(b) compounds.

(c) mixtures.



5. Classify the following substances as element, compound or mixture.

H₂O air H₂ SiO₂ cup of tea gold mercury oxide

Element	Compound	Mixture

6. Complete the following table, with either the name or symbol of the element.

ELEMENT	SYMBOL
aluminium	
	Na
nitrogen	
	B
zinc	
	Pb
iron	
	Cl
potassium	
	Ne

ELEMENT	SYMBOL
	Sn
copper	
	Si
fluorine	
	He
magnesium	
	Ag
gold	
	Hg
bromine	

7. Name the elements found in the following compounds and give the number of atoms of each.

COMPOUND	NAME AND NUMBER OF ELEMENTS PRESENT
CH ₃ COOH	
Al ₂ O ₃	
H ₂ SO ₄	

8. Complete the following table from the information presented about the Periodic Table.

	Group 1	Group 2	
Period 1	1.0 H Hydrogen 1		Relative atomic mass Symbol Atomic number
Period 2	6.9 Li Lithium 3	9.0 Be Beryllium 4	
Period 3	23.0 Na Sodium 11	24.3 Mg Magnesium 12	

Element	Symbol	Atomic number	Atomic mass	Number of protons	Number of electrons	Number of neutrons
Hydrogen						
Lithium						
Beryllium						
Sodium						
Magnesium						